

Annual Report: Project Falcon

Project Falcon began in early 2024 as an internal initiative to modernize the company's customer support pipeline. The project was led by a small cross-functional team spanning engineering, design, and operations. Over the course of the year, the team shipped three major releases, each targeting a different bottleneck in the support workflow. The first release focused on ticket routing, the second on response templates, and the third on automated escalation. Early results were promising: average response time dropped by thirty percent within the first two months, though this improvement plateaued once the easier cases had been optimized. The following sections detail each phase in more depth, along with the numerical results collected during each release cycle.

The routing release, shipped in March, introduced a simple rules engine that matched incoming tickets to the right support queue based on keywords in the subject line. This was later replaced by a lightweight classifier once the team observed that keyword matching failed on roughly fifteen percent of tickets, particularly ones written informally or in mixed language. The classifier improved routing accuracy from eighty-two percent to ninety-four percent, based on a manual audit of five hundred sampled tickets.

Before the classifier was introduced, misrouted tickets were a persistent source of frustration for both customers and agents. A ticket routed to the wrong queue could sit unanswered for hours before an agent noticed it belonged elsewhere, and manually re-routing tickets consumed a meaningful fraction of each agent's shift. The team's early investigation found that the original keyword-matching rules broke down most often on tickets that mixed languages, used unconventional abbreviations, or described a problem without naming the product directly, all patterns the classifier was specifically trained to recognize.

Rolling out the classifier also required changes to how the team measured success. Rather than relying solely on ticket volume per queue, the team introduced a weekly misroute audit, in which a sample of tickets from each queue was manually reviewed to confirm they had landed in the right place. This audit became one of the project's most valuable ongoing tools, surfacing edge cases the classifier handled poorly well before they became a widespread problem, and it remains part of the team's standard process for any future changes to the routing logic.

Looking back, the team identifies the decision to invest in a weekly misroute audit early on, rather than waiting until routing accuracy plateaued, as one of the project's better calls. It gave engineering a steady stream of real, labeled failure cases to learn from instead of relying on infrequent, large-batch reviews, and it gave support leadership an ongoing, concrete number to point to when explaining the project's progress to the rest of the company, well before the revenue impact discussed later in this report became measurable.

The audit process itself evolved over the course of the year. In the earliest weeks, a single engineer reviewed a small random sample of tickets by hand each Friday afternoon, noting misroutes in a shared spreadsheet. As ticket volume grew, this informal process was replaced by a lightweight internal tool that automatically pulled a stratified sample across queues, flagged tickets where the classifier's confidence score had been low, and let reviewers tag outcomes with a single click, cutting the weekly review time roughly in half while covering a broader and more representative slice of traffic.

Release Metrics

The table below summarizes the key metrics tracked across the three releases. Note that the escalation release, while technically successful, introduced a temporary increase in false-positive escalations during its first two weeks, which the team resolved by tightening the confidence threshold used by the escalation model.

Release	Month	Metric Improved	Change
Routing	March	Routing Accuracy	+12%
Templates	July	Avg. Response Time	-18%
Escalation	November	Escalation Precision	+9%

As shown above, the templates release had the largest single impact on response time, largely because it reduced the amount of manual typing required for common request types. Support agents reported high satisfaction with the new template library, though a few noted that overly rigid templates occasionally felt impersonal to customers.

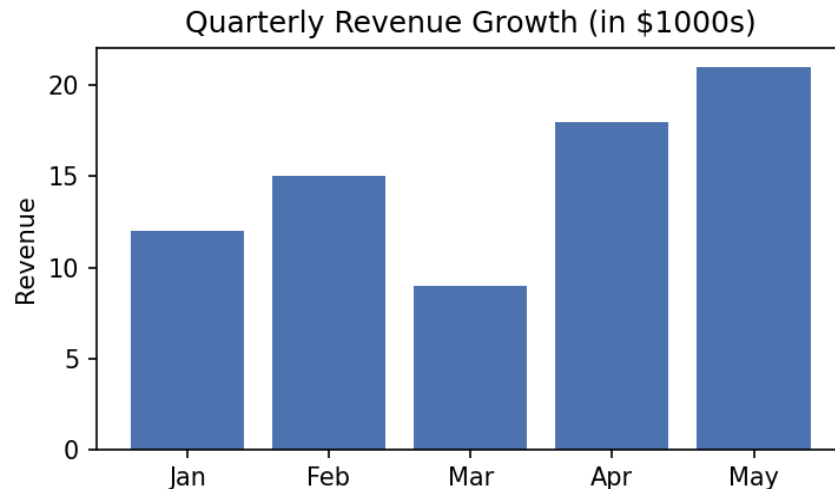
The template library itself went through several rounds of revision before reaching its current form. The first version, drafted primarily by the support operations lead, was criticized by agents for sounding too formal and for failing to cover a number of common but awkward situations, such as apologizing for a repeated issue or explaining a delay caused by a dependency on another team. A second round of templates, written collaboratively with a group of senior agents and reviewed against several months of real ticket transcripts, addressed most of these complaints and became the version deployed company-wide.

The escalation release, shipped last, proved the most technically demanding of the three. Unlike routing and templates, which primarily affected how tickets were organized and answered, escalation required the model to make a judgment call about ticket severity with limited information, often before an agent had exchanged more than one or two messages with the customer. The team addressed this by combining the model's confidence score with a small set of hard-coded override rules for known high-severity keywords, a hybrid approach that proved more reliable in production than either method alone.

One unresolved debate within the team is whether the override rules used for high-severity keywords should eventually be phased out in favor of a purely model-driven approach as the escalation model accumulates more production data. Proponents of phasing them out argue that hard-coded rules are brittle and will eventually miss phrasing the original list did not anticipate; proponents of keeping them argue that a small, human-auditable safety net is worth the added complexity for a system that directly affects how quickly a distressed customer reaches a human agent.

Revenue Impact

Beyond support efficiency, Project Falcon also had a measurable effect on revenue, primarily through improved customer retention. The chart below shows revenue growth across the five months following the first release, measured in thousands of dollars. Growth was not perfectly linear: March saw a dip, which the team attributes to a seasonal slowdown rather than any regression caused by the routing release.



The finance team's independent analysis, conducted separately from the support team's own tracking, attributed roughly two-thirds of the observed revenue growth to reduced customer churn among accounts that had previously filed at least one support ticket, with the remainder attributed to modest gains in expansion revenue from existing customers whose support experience appeared to correlate with willingness to upgrade their plan. While this analysis cannot fully rule out other contributing factors active over the same period, the correlation was strong enough that leadership approved continued investment in the support pipeline for the following fiscal year.

Looking ahead, the team plans to extend the escalation model to cover billing-related tickets in the next fiscal year, and is exploring whether the same classifier architecture used for routing could be reused for automatic ticket prioritization. No firm timeline has been set for this next phase as of this writing, though early scoping discussions suggest it would require a comparable level of effort to the original routing release, given the need to collect and label a new set of training examples specific to billing-related language and terminology.

The team also plans to publish an internal retrospective covering lessons learned across all three releases, with a particular focus on how the project's scope and success metrics evolved between the initial proposal and the final shipped system. Early drafts of the retrospective note that the original proposal did not mention customer retention or revenue at all, framing the project purely in terms of internal efficiency; the connection to revenue only became apparent once the finance team's analysis was completed well after the third release had already shipped.